

Homework 13

Connor Mooneyhan

April 29, 2026

1. Let H be a Hilbert space and define $T : H \rightarrow H'$ as $z \mapsto \langle \cdot, z \rangle$. Show that:
 - (1) T is isometric, bijective and conjugate linear: $T(ax + by) = \bar{a}T(x) + \bar{b}T(y)$.
 - (2) H' is a Hilbert space with inner product $\langle T(x), T(y) \rangle = \overline{\langle x, y \rangle} = \langle y, x \rangle$. Therefore, if H is a real Hilbert space then it is isomorphic to H' .
 - (3) Any complex Hilbert space H is isomorphic to $(H')'$.
2. Use Zorn's lemma to show that for any subspace Y of a vector space X , any Hamel basis of Y can be extended to a Hamel basis of X .
3. Let $X = \mathbb{R}^2$ be equipped with the box norm. Let $u = \langle 1, 0 \rangle$ and T be the linear functional on $\text{span}(u)$ satisfying $T(u) = 1$.
 - (1) Show that for any $M > 0$ there exists $v \in \mathbb{R}^2$, linearly independent with u , such that the extension of T determined by $T(v) = 0$ satisfies $\|T\| \geq M$.
 - (2) Find a value of c such that the extension of T determined by $T(-1, 1) = c$ satisfies $\|T\| = 1$. Is the choice of c unique?